Department of Computer Systems Engineering
Faculty of Computing
Sri Lanka Institute of Information Technology

IT3010 : Network Desing and Management
Year 3 – Semester I
2026

Assignment Title	Virtualized Network Infrastructure Deployment & Service Configuration (Individual Assignment)
Learning outcomes covered	LO5: Evaluatea network design along with a client/server network programming based on TCPand IP protocol
Assignment Mode	Two parts 1) Take home part for the environmental setup 2) Lab assessment + viva
Maximum Marks	100
Contribution to the Final Grade	30%
Date published	25 February 2026
Deadline for submissions	12th April 2026
Mode of Submission	Report and work submitted online via courseweb

Assignment description

This is an individual assignment designed to evaluate the student's ability to design and implement a virtualized network infrastructure with essential network services and a socket-based client–server system.

All students must use a virtualization environment (VMware / VirtualBox / etc.) to create a Linux based Virtual Machine (VM). *<<Students can select their choice of Linux OS.>>*

1. Virtual Infrastructure Requirements

Each student must configure their VM network using their Registration Number in order to ensure that every student develops a unique network infrastructure.

1.1. Virtual Network Adapter Configuration

Students must create the VM network address as follows:
Remove:

- First two letters of the registration number
- Last two digits of the registration number

Then convert the remaining digits into:

XX.XX.XX.0/24

Example:
Registration Number: IT12345678
Network Address : 12.34.56.0/24

1.2. Hostname Creation

Hostname must be created using:

<casual_name><last_four_digits_of_registration>

Example:
Student Name: Hettiarachchi Kamal
Registration Number: IT12345678

Hostname: kamal5678

2. Component 01 – Take Home Assessment

Students must:

2.1. DNS Server Configuration

Configure and deploy a working DNS server within the VM using the domain:

<registration_number>.lk

Example:
IT12345678.lk

2.2. Mail Server Configuration

Students must configure:

- A working Mail Server within the same VM
- A working Mail Client
- Internal mail communication capability

<<Students can select their choice of mail server which runs in Linux based.>>

2.3. Client–Server Application Development

Using **C Programming Language only**, students must implement;

- Iterative TCP Server
- Client Application

When client connects;

- Server must send **current system date and time**
- Client must display received information via terminal

Both client and server must:

- Log each session initiated
- Record socket details (IP address & port number)

2.4. Report Submission

Students must:

- Take screenshots of:
  - DNS configuration steps
  - Mail server configuration
  - Working demonstration
  - Client–server execution
- Provide brief descriptions for each step.
- Use the **CourseWeb front page template** as the report cover page.
- Upload the report before the deadline 12th April 2026 11.59 PM.

3. Component 02 – Lab Assessment

- Students must bring the fully functional Take-Home implementation (Component 01) to the lab session.
- Lab assignment assessments will informed accordingly via courseweb.
- Students **MUST** attend the allocated lab session on the mentioned date and time according to the schedule which will be published via courseweb.
- At the end of the assessment period a viva session will be carried out.
- Students should be ready to demonstrate any component of the assignment for the evaluations.
- Further instructions on the assessment will be given at the lab session.

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Marking Scheme

Criteria	Weight	Excellent (80–100%)	Good (65–79%)	Satisfactory (50–64%)	Poor (35–49%)	Very Poor (0–34%)
VM Infrastructure Setup & Unique Network Configuration	10	Fully correct addressing & hostname based on registration number	Minor configuration errors	Acceptable setup with some issues	Incorrect addressing	No working setup
DNS & Mail Server Deployment (Take-Home)	20	Fully functional DNS & internal Mail communication	Minor functional issues	Basic service working	Incomplete configuration	Not working
Client–Server Program & Logging (Take-Home)	10	Fully functional with correct session logs	Minor errors	Basic communication working	Incomplete implementation	Not working
Lab Assessment – DNS Entries & Client-Server Modification	30	Forward & Reverse DNS correctly configured and program modification fully functional	Minor issues	Basic modification working	Partial implementation	Unable to complete
Viva Performance & Demonstration	30	Clearly explains all components and demonstrates services confidently	Good understanding with minor gaps	Basic explanation	Limited understanding	Unable to explain or demonstrate